

Ai Assisted Coding

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Batch: 45

Assignment:11.3

Task 1: Smart Contact Manager (Arrays & Linked Lists)

Scenario: SR University's student club requires a simple Contact Manager Application to store members' names and phone numbers. The system should support efficient addition, searching, and deletion of contacts.

Tasks

1. Implement the contact manager using arrays (lists).
2. Implement the same functionality using a linked list for dynamic memory allocation.
3. Implement the following operations in both approaches:
 - o Add a contact
 - o Search for a contact
 - o Delete a contact
4. Use GitHub Copilot to assist in generating search and delete methods.
5. Compare array vs. linked list approaches with respect to:
 - o Insertion efficiency
 - o Deletion efficiency

Prompt:

Generate a python code by using DSA. SR University's student club requires a simple Contact Manager Application to store members' names and phone numbers. The system should support efficient addition, searching, and deletion of contacts. Implement using arrays and linked list and also include in arrays add a contact,search for a contact, delete for a contact.

Code:

```
class ArrayContactManager:
```

```
    def __init__(self):
```

```
        self.contacts = []
```

```

def add_contact(self, name, phone):
    self.contacts.append({'name': name.strip(), 'phone': phone.strip()})

def search_contact(self, name):
    key = name.strip().lower()
    return [c for c in self.contacts if c['name'].lower() == key]

def delete_contact(self, name):
    key = name.strip().lower()
    before = len(self.contacts)
    self.contacts = [c for c in self.contacts if c['name'].lower() != key]
    return before - len(self.contacts)

```

```

class Node:

```

```

    def __init__(self, name, phone):
        self.name = name.strip()
        self.phone = phone.strip()
        self.next = None

```

```

class LinkedListContactManager:

```

```

    def __init__(self):
        self.head = None

    def add_contact(self, name, phone):
        new_node = Node(name, phone)
        if not self.head:
            self.head = new_node
            return
        curr = self.head
        while curr.next:
            curr = curr.next
        curr.next = new_node

    def search_contact(self, name):
        key = name.strip().lower()

```

```

out = []
curr = self.head
while curr:
    if curr.name.lower() == key:
        out.append({'name': curr.name, 'phone': curr.phone})
        curr = curr.next
    return out

def delete_contact(self, name):
    key = name.strip().lower()
    dummy = Node("", "")
    dummy.next = self.head
    prev, curr = dummy, self.head
    deleted = 0
    while curr:
        if curr.name.lower() == key:
            prev.next = curr.next
            deleted += 1
        else:
            prev = curr
        curr = curr.next
    self.head = dummy.next
    return deleted

if __name__ == '__main__':
    a = ArrayContactManager()
    a.add_contact('Alice', '111')
    a.add_contact('Bob', '222')
    print('Array search Bob:', a.search_contact('Bob'))
    print('Array delete Bob:', a.delete_contact('Bob'), 'left', a.contacts)
    l = LinkedListContactManager()

```

```

l.add_contact('Alice', '111')

l.add_contact('Bob', '222')

print('Linked search Bob:', l.search_contact('Bob'))

print('Linked delete Bob:', l.delete_contact('Bob'))

print('Linked search Bob after:', l.search_contact('Bob'))

```

Output:

```

Array search Bob: [{'name': 'Bob', 'phone': '222'}]
Array delete Bob: 1 left [{'name': 'Alice', 'phone': '111'}]
Linked search Bob: [{'name': 'Bob', 'phone': '222'}]
Linked delete Bob: 1
Linked search Bob after: []

```

Explanation:

In this code we can code and output we can see that we have entered the contacts, array search bob, array delete bob and so on.

TASK:02 Book Search System (Queues & Priority Queues)

Scenario: The SRU Library manages book borrow requests. Students and faculty submit requests, but faculty requests must be prioritized over student requests.

Tasks

1. Implement a Queue (FIFO) to manage book requests.
2. Extend the system to a Priority Queue, prioritizing faculty requests.
3. Use GitHub Copilot to assist in generating: enqueue() method , dequeue() method
4. Test the system with a mix of student and faculty requests.

Prompt:

Generate a python code for SRU Library manages book borrow requests. Students and faculty submit requests, but faculty requests must be prioritized over student requests. Also include Implement a Queue (FIFO) to manage book requests. Extend the system to a Priority Queue, prioritizing faculty requests. enqueue() method , dequeue() method Test the system with a mix of student and faculty requests. In data structures.

Code:

```

import heapq

from collections import deque

from itertools import count

class Req:

```

```

def __init__(self,id,type,title):
    self.id, self.type, self.title = id,type.lower(),title
def __repr__(self): return f"{self.type[0].upper()}{self.id}:{self.title}"
class FIFO:
    def __init__(self): self.q = deque()
    def enqueue(self,r): self.q.append(r)
    def dequeue(self): return self.q.popleft() if self.q else None
class PQueue:
    def __init__(self): self.h, self.c = [], count()
    def enqueue(self,r):
        p=0 if r.type=="faculty" else 1
        heapq.heappush(self.h,(p,next(self.c),r))
    def dequeue(self): return heapq.heappop(self.h)[2] if self.h else None
reqs = [Req("1","student","DS"), Req("1","faculty","AI"),
        Req("2","student","Calc"), Req("2","faculty","ML"), Req("3","student","Net")]
f, pq = FIFO(), PQueue()
for r in reqs:
    f.enqueue(r); pq.enqueue(r)
fifo_out = [f.dequeue() for _ in range(len(reqs))]
pri_out = [pq.dequeue() for _ in range(len(reqs))]
print(f"{'FIFO':<20}{'PRIORITY':<20}")
print("-"*40)
for a,b in zip(fifo_out, pri_out):
    print(f"{str(a):<20}{str(b):<20}")

```

Output:

```

FIFO                PRIORITY
-----
S1:DS                F1:AI
F1:AI                F2:ML
S2:Calc              S1:DS
F2:ML                S2:Calc
S3:Net               S3:Net

```

Explanation:

In this code we can see that which faculty is selecting which student and we can also see that which subject they are choosing.

Task 3: Emergency Help Desk (Stack Implementation)**Scenario**

SR University's IT Help Desk receives technical support tickets from students and staff. While tickets are received sequentially, issue escalation follows a Last-In, First-Out (LIFO) approach.

Tasks

1. Implement a Stack to manage support tickets.
2. Provide the following operations:
 - o push(ticket)
 - o pop()
 - o peek()
3. Simulate at least five tickets being raised and resolved.
4. Use GitHub Copilot to suggest additional stack operations such as:
 - o Checking whether the stack is empty
 - o Checking whether the stack is full (if applicable)

Prompt:

Generate a python code for SR University's IT Help Desk receives technical support tickets from students and staff. While tickets are received sequentially, issue escalation follows a Last-In, First-Out (LIFO) approach. Implement a Stack to manage support tickets. Provide the following operations: push(ticket), pop(), peek(). Simulate at least five tickets being raised and resolved. Use GitHub Copilot to suggest additional stack operations such as: Checking whether the stack is empty, Checking whether the stack is full (if applicable) By using data structures.

Code:

```
class Ticket:
```

```
    def __init__(self, i, u, iss):  
        self.i = i  
        self.u = u  
        self.iss = iss
```

```

def __repr__(self):
    return f"{self.i}:{self.u}:{self.iss}"

class Stack:
    def __init__(self, c=None):
        self.s = []
        self.c = c

    def push(self, t):
        if self.c is not None and len(self.s) >= self.c:
            raise OverflowError("Stack full")
        self.s.append(t)

    def pop(self):
        return self.s.pop() if self.s else None

    def peek(self):
        return self.s[-1] if self.s else None

    def is_empty(self):
        return not self.s

    def is_full(self):
        return self.c is not None and len(self.s) == self.c

st = Stack(10)

for x in [
    ("T1", "student", "Wi-Fi down"),
    ("T2", "staff", "Printer error"),
    ("T3", "student", "Password reset"),
    ("T4", "staff", "VPN timeout"),
    ("T5", "student", "Email sync")]:
    st.push(Ticket(*x))

print("peek:", st.peek())

print("empty:", st.is_empty(), "full:", st.is_full())

while not st.is_empty():

```

```
print("resolved:", st.pop())
```

Output:

```
peek: T5:student:Email sync
empty: False full: False
resolved: T5:student:Email sync
resolved: T4:staff:VPN timeout
resolved: T3:student:Password reset
resolved: T2:staff:Printer error
resolved: T1:student:Wi-Fi down
```

Expalnation:

Stack class manages helpdesk tickets in LIFO order with `push`, `pop`, `peek`, `is\_empty`, `is\_full` and resolves newest tickets first.

Task 4: Hash Table Objective.To implement a Hash Table and understand collision handling.

Task Description

Use AI to generate a hash table with:

- Insert
- Search
- Delete

Starter Code

```
class HashTable:
```

```
pass
```

Prompt:

Generate a python code for : Hash Table Objective To implement a Hash Table and understand collision handling.

Task DescriptionUse AI to generate a hash table with: Insert , Search, Delete

Starter Code

```
class HashTable:
```

```
pass
```

Code:

```
class HashTable:
```

```
    def __init__(self, size=7):
```

```
        self.size = size
```



```

        self.buckets = [[] for _ in range(size)]
def _idx(self, key):
    return hash(key) % self.size
def insert(self, key, value):
    idx = self._idx(key)
    bucket = self.buckets[idx]
    for i, (k, _) in enumerate(bucket):
        if k == key:
            bucket[i] = (key, value)
            return
    bucket.append((key, value))
def search(self, key):
    idx = self._idx(key)
    for k, v in self.buckets[idx]:
        if k == key:
            return v
    return None
def delete(self, key):
    idx = self._idx(key)
    bucket = self.buckets[idx]
    for i, (k, _) in enumerate(bucket):
        if k == key:
            bucket.pop(i)
            return True
    return False

h = HashTable()
h.insert("a", 1)
h.insert("b", 2)
h.insert("a", 3 )

```

```
print(h.search("a"))
print(h.search("b"))
print(h.delete("b"))
print(h.search("b"))
```

Output:

```
3
2
True
None
```

Explanation:

HashTable uses fixed-size buckets + chaining so `insert/search/delete` map keys to bucket by `hash(key)%size` and handle collisions in per-bucket list.

Task 5: Real-Time Application Challenge

Scenario

Design a Campus Resource Management System with the following

features:

- Student Attendance Tracking
- Event Registration System
- Library Book Borrowing
- Bus Scheduling System
- Cafeteria Order Queue

Student Tasks

1. Choose the most appropriate data structure for each feature.
2. Justify your choice in 2–3 sentences.
3. Implement one selected feature using AI-assisted code generation.

Prompt:

Generate a python code for Real-Time Application Challenge

Design a Campus Resource Management System with the following

features:Student Attendance Tracking,Event Registration System,Library Book Borrowing,Bus Scheduling System,Cafeteria Order Queue

Student Tasks:Choose the most appropriate data structure for each feature.

Justify your choice in 2–3 sentences. Implement one selected feature using AI-assisted code generation.

Code:

```
from collections import deque
from dataclasses import dataclass
@dataclass
class Order:
    id: int
    student: str
    items: list
    def __str__(self):
        return f"#{self.id} {self.student}: {' '.join(self.items)}"
class CafeteriaQueue:
    def __init__(self):
        self._queue = deque()
        self._next_id = 1
    def place_order(self, student_name, items):
        order = Order(self._next_id, student_name, items)
        self._next_id += 1
        self._queue.append(order)
        print("Order placed:", order)
        return order.id
    def serve_order(self):
        if not self._queue:
            print("No pending orders.")
            return None
        order = self._queue.popleft()
        print("Order served:", order)
        return order
```

```

def pending_orders(self):
    return list(self._queue)

if __name__ == "__main__":
    q = CafeteriaQueue()
    q.place_order("Aisha", ["Sandwich", "Juice"])
    q.place_order("Ravi", ["Samosa", "Tea"])
    q.place_order("Mina", ["Pasta", "Water"])
    q.place_order("Anil", ["Burger", "Soda"])
    q.place_order("Sara", ["Salad", "Smoothie"])
    print("Pending:", q.pending_orders())
    q.serve_order()
    q.serve_order()
    print("Pending after serving two:", q.pending_orders())

```

Output:

```

Order placed: #1 Aisha: Sandwich, Juice
Order placed: #2 Ravi: Samosa, Tea
Order placed: #3 Mina: Pasta, Water
Order placed: #4 Anil: Burger, Soda
Order placed: #5 Sara: Salad, Smoothie
Pending: [Order(id=1, student='Aisha', items=['Sandwich', 'Juice']), Order(id=2, student='Ravi', items=['Samosa', 'Tea']), Order(id=3, student='Mina', items=['Pasta', 'Water']), Order(id=4, student='Anil', items=['Burger', 'Soda']), Order(id=5, student='Sara', items=['Salad', 'Smoothie'])]
Order served: #1 Aisha: Sandwich, Juice
Order served: #2 Ravi: Samosa, Tea
Pending after serving two: [Order(id=3, student='Mina', items=['Pasta', 'Water']), Order(id=4, student='Anil', items=['Burger', 'Soda']), Order(id=5, student='Sara', items=['Salad', 'Smoothie'])]

```

Expalantion:

Cafeteria queue uses `deque` for FIFO order so orders are placed and served in real-world sequence.